3. Estimation, confidence intervals and tests

What students need to learn:

Concepts of standard error, estimator, bias.	The sample mean, $\bar{x}$, and the sample variance, $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$, as unbiased estimates of the corresponding population parameters.

The distribution of the sample mean $\overline{X}$.	$\overline{X}$ has mean μ and variance $\dfrac{\sigma^2}{n}$. If $X \sim \mathrm{N}(\mu, \sigma^2)$ then $\overline{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{n}\right)$. No proofs required.
Concept of a confidence interval and its interpretation.	Link with hypothesis tests.
Confidence limits for a Normal mean, with variance known.	Students will be expected to know how to apply the Normal distribution and use the standard error and obtain confidence intervals for the mean, rather than be concerned with any theoretical derivations.
Hypothesis tests for the mean of a Normal distribution with variance known.	Use of $\dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0, 1)$.
The Central Limit theorem.	Students will be expected to know how to apply the Central Limit theorem to contexts and distributions within the S3 content.
Use of Central Limit theorem to extend hypothesis tests and confidence intervals to samples from non-Normal distributions. Use of large sample results to extend to the case in which the variance is unknown.	$\dfrac{\overline{X} - \mu}{S/\sqrt{n}}$ can be treated as $\mathrm{N}(0, 1)$ when n is large. A knowledge of the t-distribution is not required.
Hypothesis test for the difference between the means of two Normal distributions with variances known.	Use of $\dfrac{(\overline{X} - \overline{Y}) - (\mu_x - \mu_y)}{\sqrt{\dfrac{\sigma_x^2}{n_x} + \dfrac{\sigma_y^2}{n_y}}} \sim \mathrm{N}(0, 1)$.
Use of large sample results to extend to the case in which the population variances are unknown.	Use of $\dfrac{(\overline{X} - \overline{Y}) - (\mu_x - \mu_y)}{\sqrt{\dfrac{S_x^2}{n_x} + \dfrac{S_y^2}{n_y}}} \sim \mathrm{N}(0, 1)$. A knowledge of the t-distribution is not required.